MAIA

Intelligent Monitoring & Auto-Healing Platform

Your AI Monitoring Assistant for Intelligent Operations

System Specification Document
Version 2.0 · May 2026

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1 Executive Summary

MAIA is an intelligent monitoring and auto-healing platform for automated job pipelines (DTSX/SSIS). It continuously watches for failures, classifies them by type, generates AI-driven fix recommendations, and either resolves them automatically or routes them to an operator for approval — all through a modern web-based dashboard.

■ Faster Detection	■ Less Manual Work	■ Full Audit Trail	■ Self-Service Config
Seconds vs. hours of manual log review	Configurable auto-heal rules reduce toil	Every action recorded immutably	No database access needed for config

2 System Architecture

MAIA watches your pipelines, understands what went wrong, knows how to fix it, and either fixes it automatically or tells the operator exactly what to do.

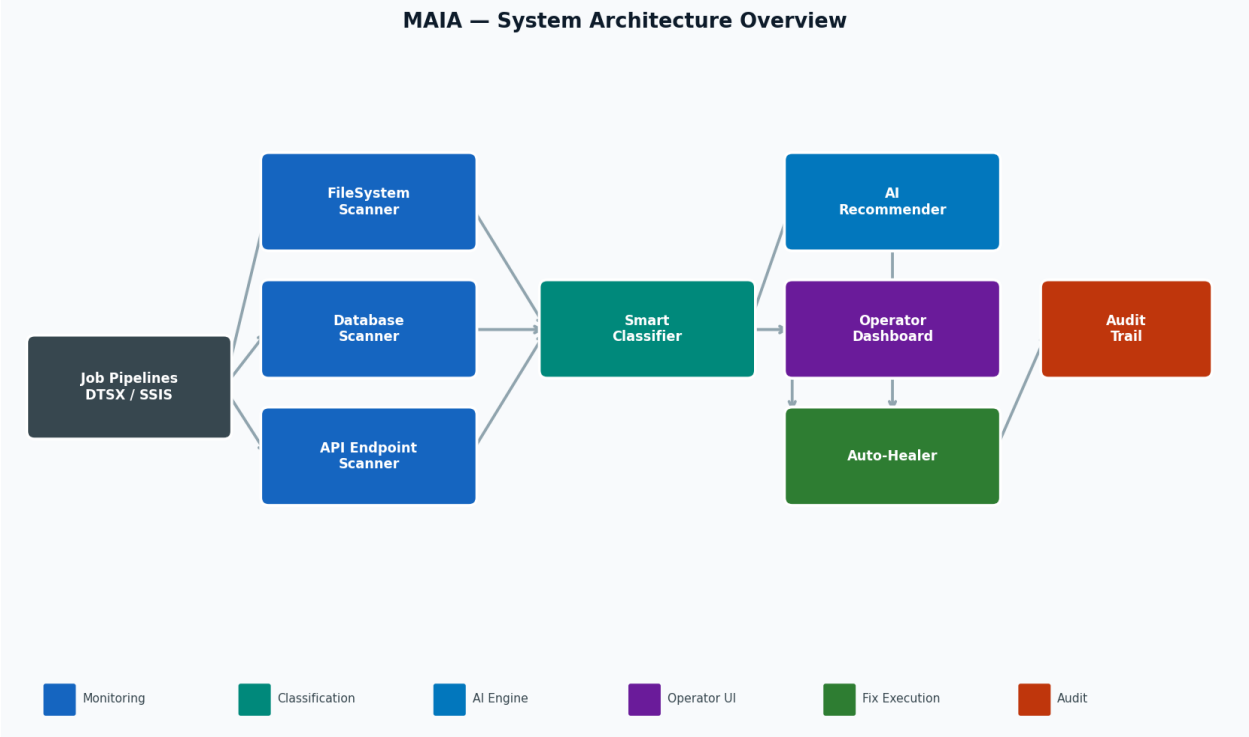

Figure 1 — MAIA System Architecture Overview

3 Key Stakeholders & User Roles

Role	Who They Are	What They Do
Operator	Data / Support Engineer	Reviews failures, approves fixes, sets auto-heal rules
Admin	Senior Engineer / Team Lead	Configures monitored jobs, classification and fix policies
System (Auto)	MAIA itself	Runs scans, classifies failures, executes auto-heal fixes
Auditor	Compliance / Management	Reviews the immutable audit log for compliance

4 Operator Workflow

The primary day-to-day workflow for an operator responding to a pipeline failure.

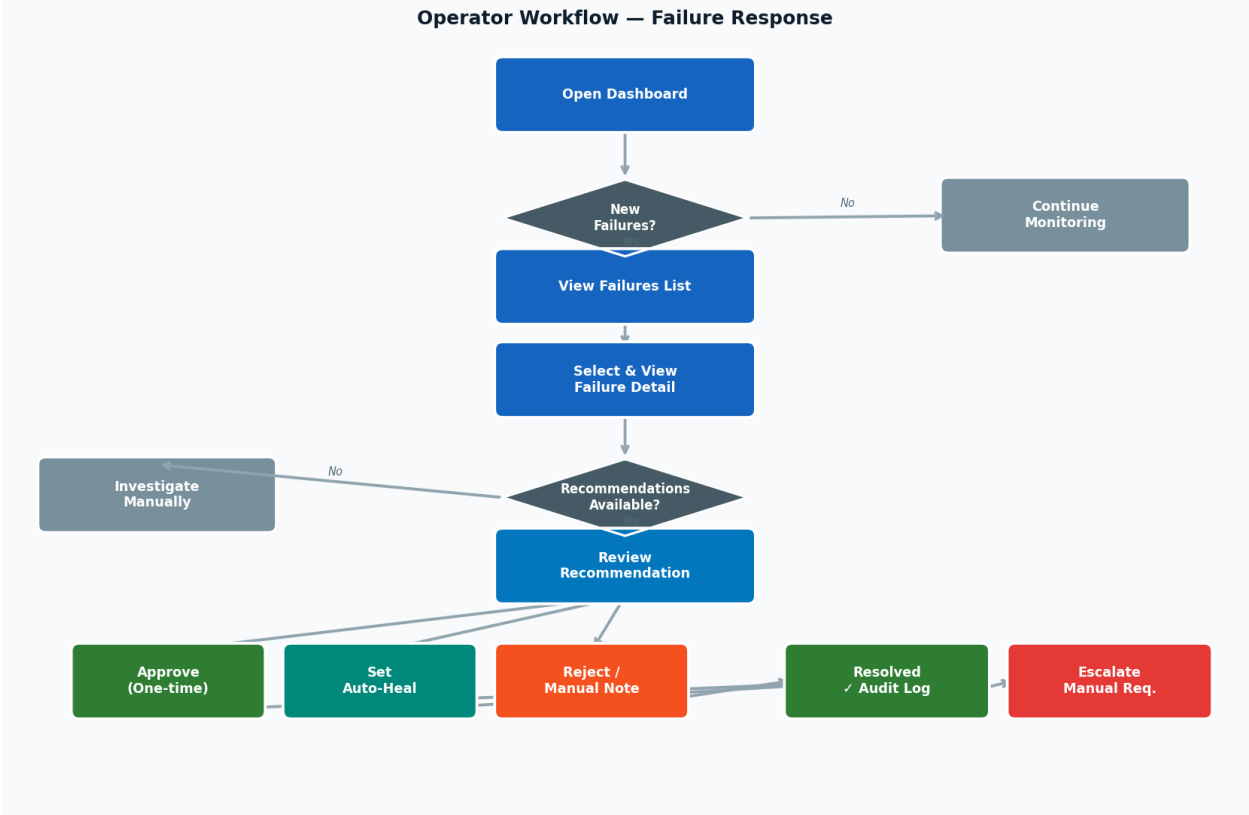

Figure 2 — Operator Failure Response Workflow

5 Auto-Heal Workflow

Once auto-heal rules are configured, MAIA resolves failures without any human intervention. The background worker runs every 60 seconds.

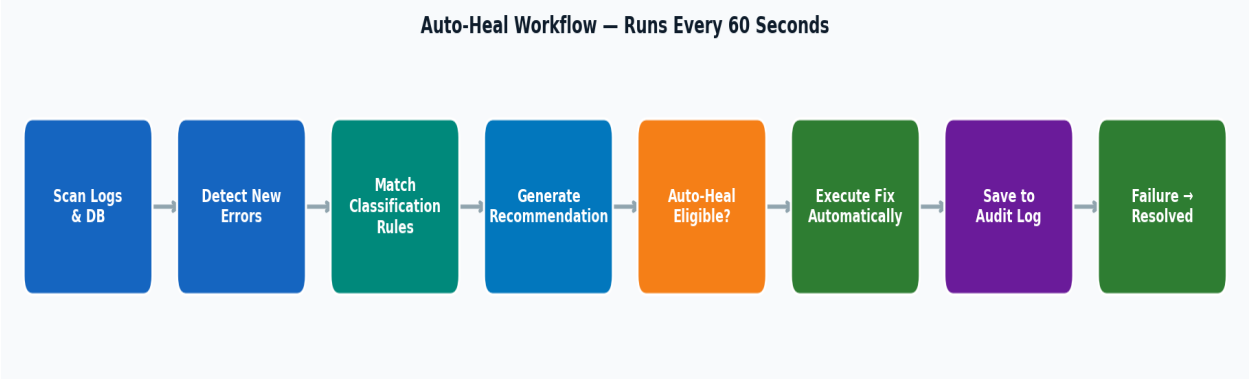

Figure 3 — Auto-Heal Background Process

Auto-Heal is Enabled When…
An operator previously approved a recommendation and toggled "Set as Auto-Heal"
A Fix Policy Rule for that error type has IsAutoHealEligible = true configured by an admin

6 Configuration Management Workflow

How an admin sets up a new job for monitoring through the web UI — no database access required.

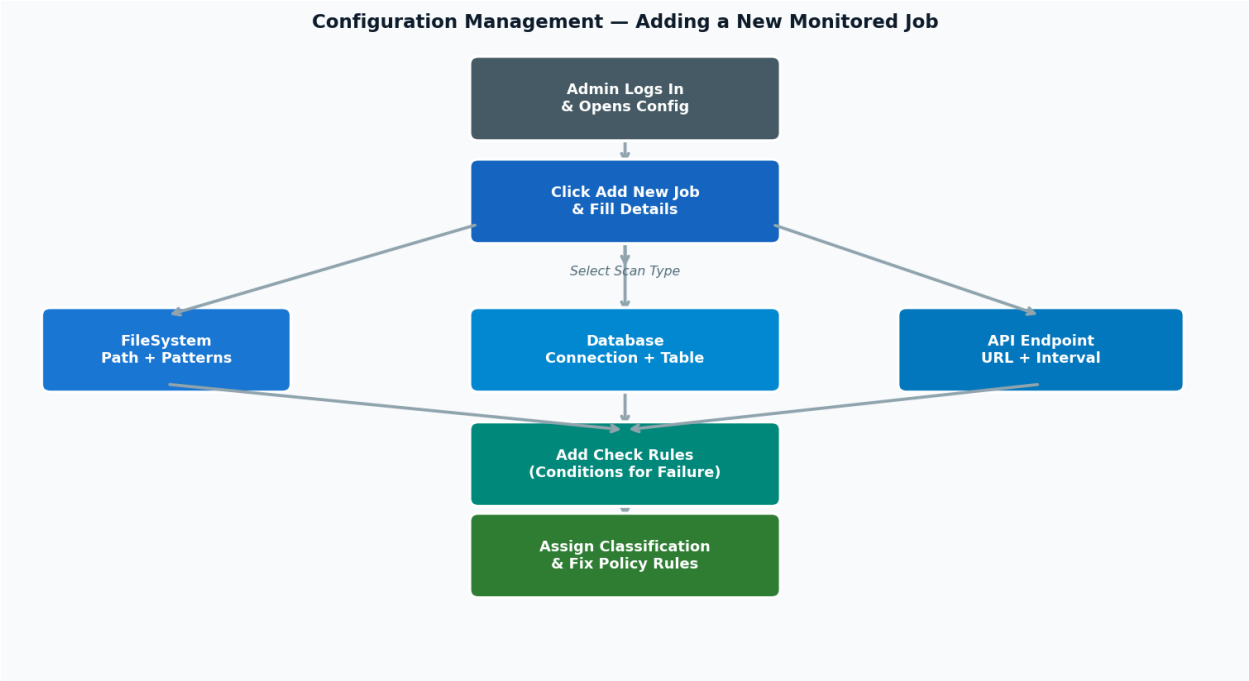

Figure 4 — New Job Configuration Workflow

7 Feature Status

Figure 5 — Feature Build Status

8 Glossary

Term	Definition
MonitoredJob	A configured pipeline that MAIA watches. Has scan type, polling interval, and rules.
ScanCheckRule	A condition evaluated during a DB scan (e.g. "row count must be > 100").
ClassificationRule	A regex pattern that assigns a JobType and ErrorType to a failure.
JobFailure	A detected failure instance — one error event from a pipeline.
AiRecommendation	A suggested fix action with confidence score and auto-heal eligibility.
FixPolicyRule	Admin-configured rule: for ErrorType X on JobType Y, run this action.
Auto-Heal	Automatic fix execution without operator approval, governed by IsAutoHealEligible.
OperatorAction	A human decision recorded — approve, reject, or set auto-heal.
FixExecutionLog	Result record of a fix attempt: success/failure, timestamp, trigger type.
AuditLog	Immutable append-only log of all significant system actions.
Watermark	Saved position (file offset or DB row ID) enabling incremental scanning.
Confidence Score	A 0.0–1.0 value indicating certainty of a classification or suggestion.
FixCategory	Broad fix category: Retry, FileRepair, DbFix, or Manual.
DTSX	File format for SQL Server Integration Services (SSIS) packages.

TECHNICAL APPENDIX

Architecture · Workers · Data Flow · Fix Engine · Scan Strategies · Database Schema · API · NFR

Appendix A Clean Architecture Layers

MAIA follows Clean Architecture — all dependencies point inward only. The Core domain has zero external dependencies; Infrastructure implements Core interfaces.

Layer	Components	Responsibility
Presentation (AIEngineAPI)	DataController · ConfigController ClassificationController · FixController PipelineController · GlobalExceptionHandler	HTTP endpoints, request routing, error formatting (RFC 7807)
Application (Use Cases)	ClassifyJobsUseCase GenerateSuggestionsUseCase ExecuteFixesUseCase DirectoryPipelineUseCase	Orchestrate domain logic, no infrastructure dependencies
Core (Domain)	Entities: JobFailure · MonitoredJob AiRecommendation · FixPolicyRule Interfaces: IFixEngine · IScanStrategy	Business rules & contracts, zero external dependencies
Infrastructure	SqlRepositories (9×) · Scan Strategies Fix Executors · RuleBasedClassifier Background Workers (4×)	Data access, external calls, implements Core interfaces

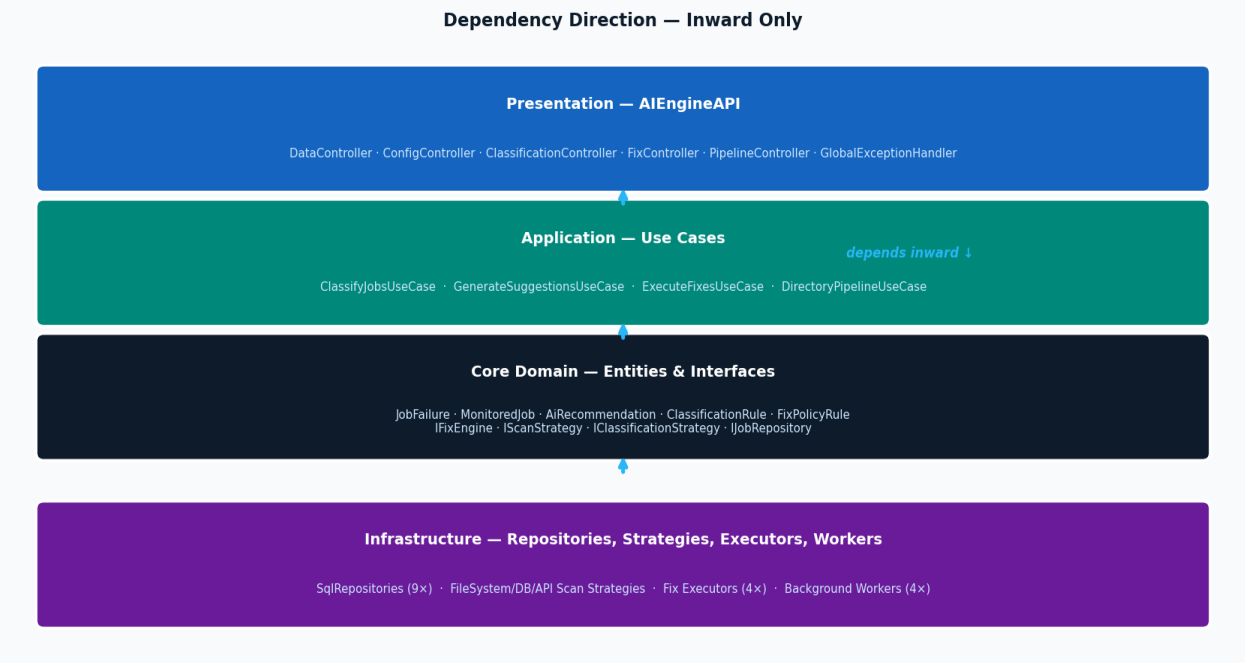

Figure A1 — Clean Architecture Layer Dependency

Appendix B Background Worker Pipeline

MAIA runs four background workers. The MonitoringWorker drives the complete pipeline every 60 seconds. The other three run the same use cases independently for targeted operations.

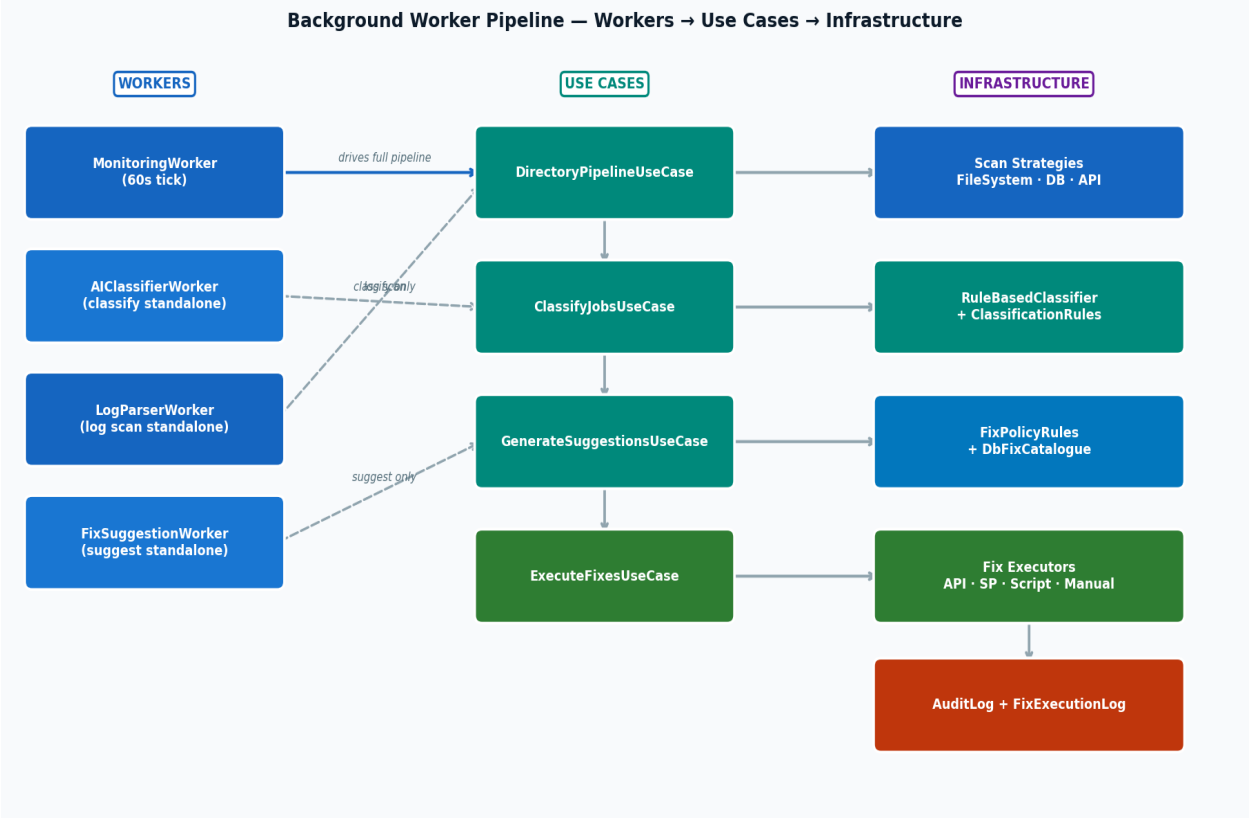

Figure B1 — Background Worker Pipeline & Use Case Orchestration

Appendix C End-to-End Data Flow

Traces the complete journey of a failure event — from pipeline detection through classification, recommendation generation, and fix execution to final resolution.

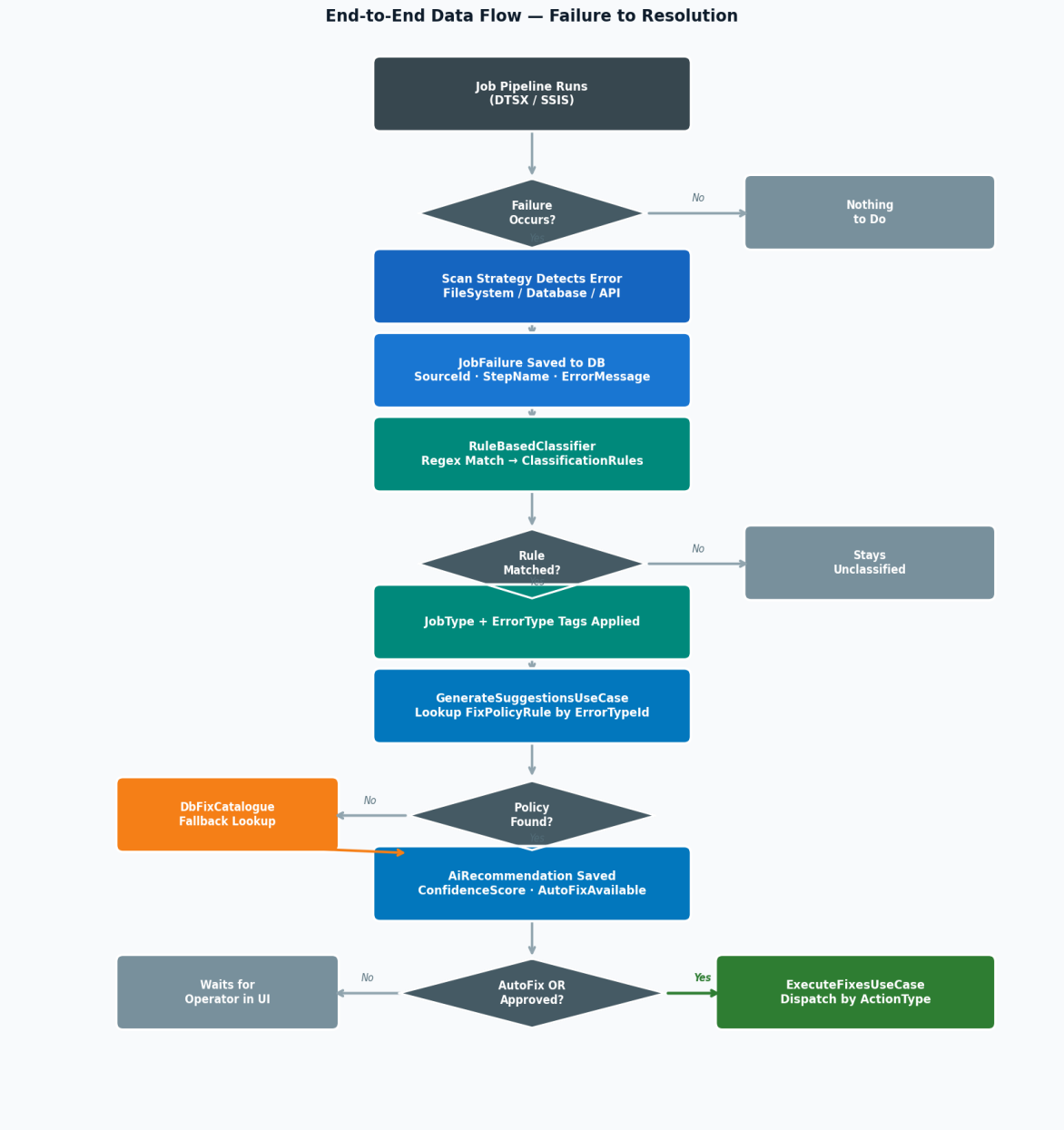

Figure C1 — End-to-End Data Flow: Pipeline Failure to Resolution

Appendix D Fix Execution Engine

When a fix is approved (manually or via auto-heal), ExecuteFixesUseCase dispatches to the appropriate executor based on the FixPolicyRule ActionType. If no policy exists, it falls back to category-based handlers.

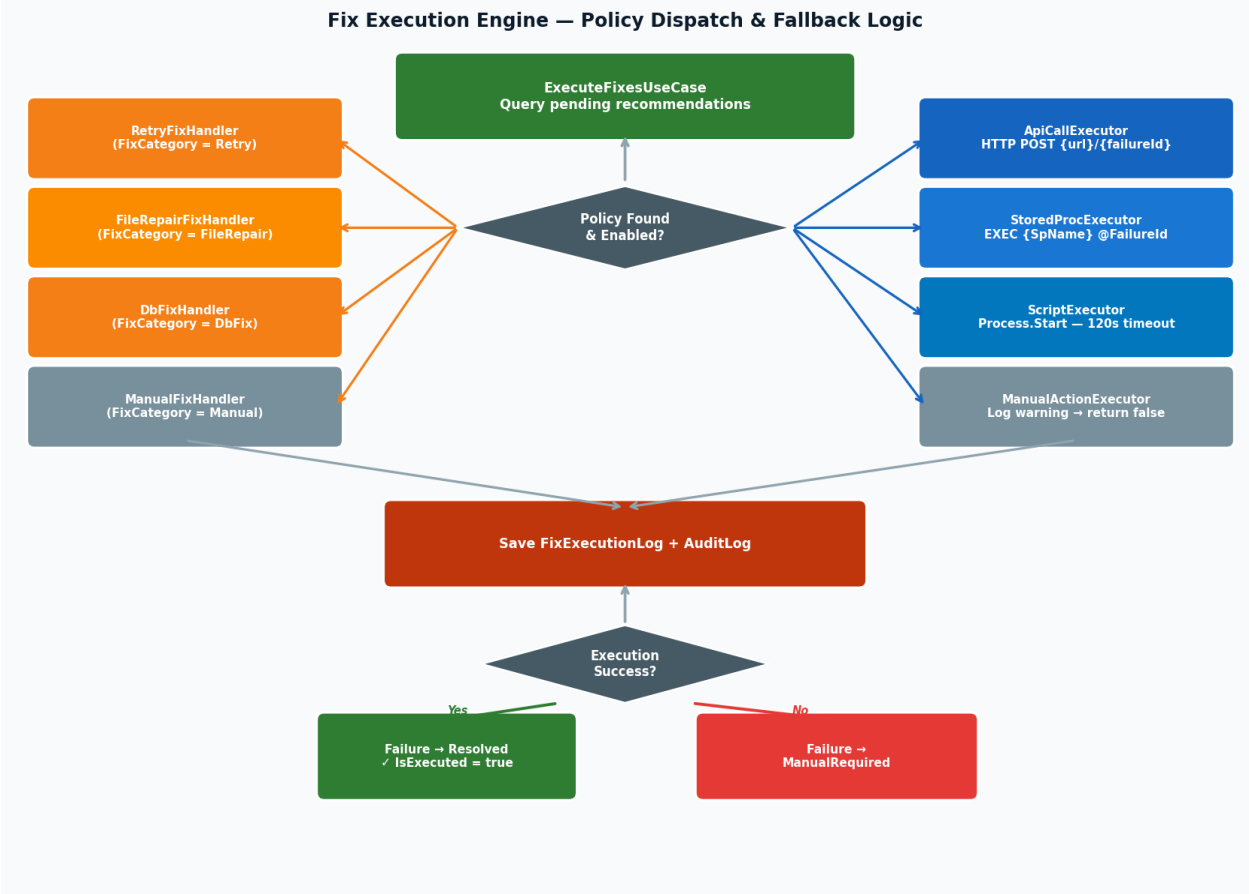

Figure D1 — Fix Execution Engine: Policy Dispatch & Fallback Handlers

Appendix E Scan Strategies

Each MonitoredJob is configured with one of three scan strategies. All strategies save discovered failures to SQL Server and use watermarks to prevent re-processing.

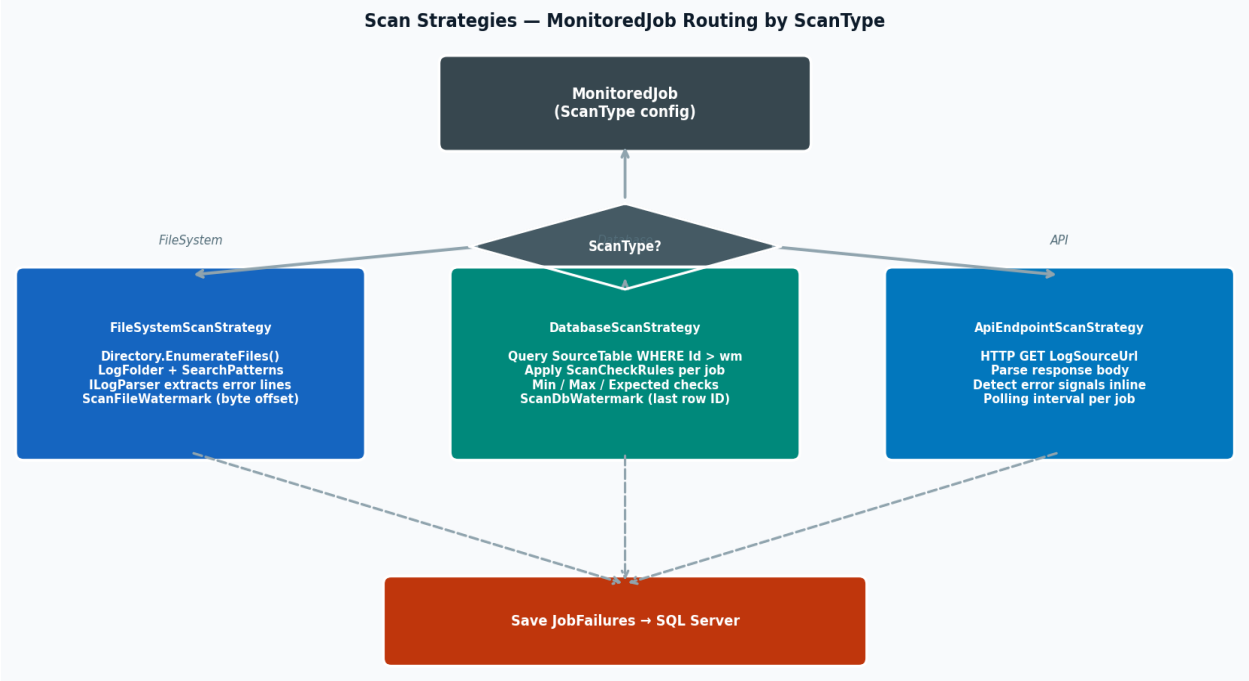

Figure E1 — Scan Strategy Selection & Watermark Tracking

Strategy	Input	Watermark	Check Rules
FileSystemScanStrategy	Log files in LogFolder (SearchPatterns e.g. *.log)	ScanFileWatermark (byte offset)	N/A — parses error lines directly
DatabaseScanStrategy	SourceTable rows WHERE Id > watermark	ScanDbWatermark (last processed row ID)	ScanCheckRules (Min/Max/Expected)
ApiEndpointScanStrategy	HTTP GET response from LogSourceUrl	Per polling interval (no persistent watermark)	Response body error signal detection

Appendix F Database Schema

MAIA uses SQL Server with 9 core tables. All primary keys are UNIQUEIDENTIFIERs. The AuditLog table is append-only. Watermarks enable incremental scanning without reprocessing.

■ Primary Key (PK)	■ Foreign Key (FK)	Regular column

Figure F1 — MAIA Database Schema & Entity Relationships

Appendix G API Endpoint Catalog

All HTTP endpoints exposed by AIEngineAPI, organized by controller.

DataController — Read-Only Dashboard Queries

Method	Endpoint	Description	Response
GET	/api/data/failures	Paginated job failures list	PagedResult<JobFailureDto>
GET	/api/data/failures/{id}/status	Failure detail + recommendations	FailureStatusDto
GET	/api/data/recommendations	Paginated AI recommendations	PagedResult<RecommendationDto>
GET	/api/data/monitored-jobs	All active monitored jobs with rules	MonitoredJobDto[]

ConfigController — CRUD Configuration

Method	Endpoint	Description	Response
CRUD	/api/config/monitored-jobs	Manage MonitoredJob entities	GET/POST/PUT/DELETE
CRUD	/api/config/scan-check-rules	Manage ScanCheckRules per job	GET/POST/PUT/DELETE
CRUD	/api/config/classification-rules	Manage ClassificationRules	GET/POST/PUT/DELETE
CRUD	/api/config/fix-policy-rules	Manage FixPolicyRules	GET/POST/PUT/DELETE

Processing Controllers — Pipeline Triggers

Method	Endpoint	Description	Response
POST	/api/classification/classify	Run classification on pending failures	200 OK
POST	/api/fix/execute-fixes	Execute approved / auto-heal fixes	200 OK
POST	/api/pipeline/run-pipeline	Run full directory pipeline	200 OK
POST	/api/logparser/parse	Parse a log file on demand	200 OK

Appendix H Non-Functional Requirements

Concern	Approach
Reliability	MonitoringWorker restarts on crash (hosted service); AuditLog is append-only
Offline Support	Rules and config stored in SQL Server; no cloud dependency at runtime
Scalability	Worker projects can run as separate services or containers
Auditability	Every fix execution writes to FixExecutionLog and immutable AuditLog
Error Handling	GlobalExceptionHandler returns RFC 7807 ProblemDetails on all exceptions
Extensibility	New scan type: implement IScanStrategy. New fix: implement IFixActionExecutor
Incremental Scanning	ScanWatermarks prevent re-processing already-seen records
Confidence Scoring	ClassificationRules carry Confidence (0.0–1.0); recommendations inherit score